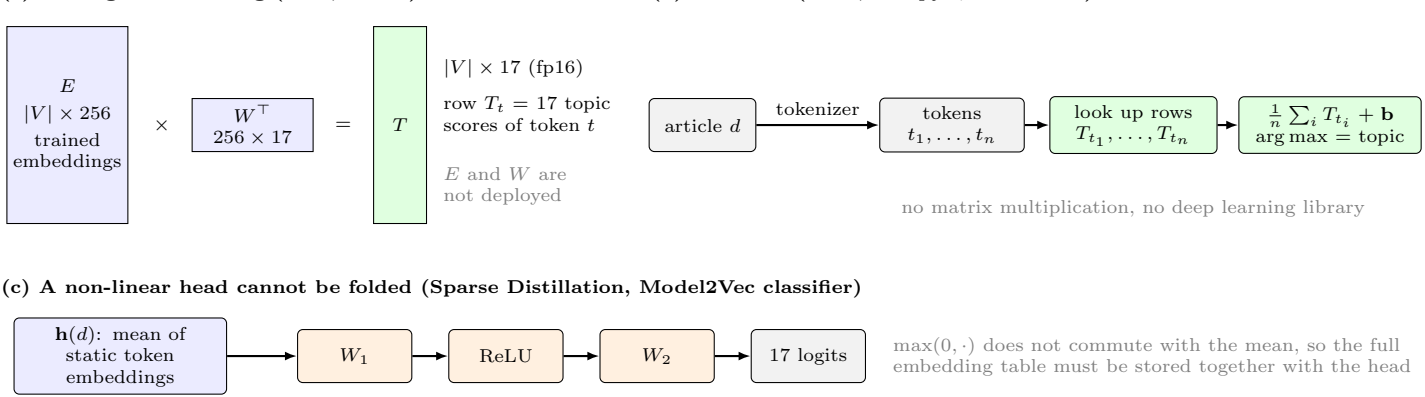

(a) Folding after training (once, offline)
E
|V| × 256
trained
embeddings
×
W⊤
256 × 17
=
T
|V| × 17 (fp16)
row T_t = 17 topic
scores of token t
E and W are
not deployed
(b) Inference (CPU, numpy + tokenizers)
article d
tokenizer
tokens
t_1, . . . , t_n
look up rows
T_{t_1}, . . . , T_{t_n}
1/n Σ_i T_{t_i} + b
arg max = topic
no matrix multiplication, no deep learning library
(c) A non-linear head cannot be folded (Sparse Distillation, Model2Vec classifier)
h(d): mean of
static token
embeddings
W_1
ReLU
W_2
17 logits
max(0, ·) does not commute with the mean, so the full
embedding table must be stored together with the head